Task: Build a Crypto Trading Dashboard

Develop a web-based trading app. The app will allow users to view real-time crypto prices, analyze trends with charts, and simulate trades for popular cryptocurrencies like Bitcoin and Ethereum. Users can also view their portfolio summary with profit/loss calculations.

Main Features

1. Crypto Price Chart

- Display a real-time price chart for a selected cryptocurrency (e.g., BTC or ETH).
- Show price movements for the past 24 hours.
- Use a line chart to plot the data.

2. Mock Trading

- Create a buy/sell interface where users can simulate trades for selected cryptocurrencies.
- Update the user's portfolio immediately after a trade (no backend required).

3. Portfolio Overview

- Display the user's current crypto holdings and transaction history.
- Calculate and display profit/loss based on the latest prices.

Use the Figma specification as a reference for design implementation. While the provided design is for a mobile app, adapt the same style to create a responsive design suitable for desktop screens.

Tech Requirements

React

- Build the app using React. Use modern React features such as hooks.

State Management

- Use Redux (or another state management library) to handle the app's data, such as prices, trades, and portfolio information.

Charting

- Use a charting library like Recharts or Chart.js.

Data Source

- Use a free API like CoinGecko, CoinCap, or Binance API.

Error Handling

- Gracefully handle errors, such as when API calls fail or return invalid data.

Styling

- Use your preferred CSS framework or library (e.g., Tailwind CSS, Bootstrap, or plain CSS).

Extra Points

1. **Testing**
    a. Write unit tests for key components using a testing library such as Jest or React Testing Library.
2. **Performance Optimization**
    a. Implement optimizations like debouncing the search input and lazy loading components.
3. **Real-Time Data Updates**
    a. Fetch crypto prices periodically (e.g., every 10 seconds) to update the UI.

Development Practices

- Use Git for version control, and ensure meaningful commit messages.
- Write clean, modular, and reusable code.
- Keep it as simple as the nature of this problem allows it. Do not over-engineer.

Submission Guidelines

Share the GitHub repository link with a README file containing:

- Instructions for setting up and running the app locally.